Molecules on Mercury have speed higher than the escape speed, thus they are fast enough to escape from Mercury. Mercury has no atmosphere.

6

(a) By conservation of energy,

Thermal Energy gained by ice = Thermal energy lost by water

$0.024(2.1\text{x}10^3)(15) + 0.024(330\text{x}10^3) = 0.20(4.2\text{x}10^3)(28\text{-}T)$

$T = 17.7\ {}^\circ C$

(b)(i) PV/T = constant

$T = (6.5 \times 10^6 \times 30 \times 300)/(1.1 \times 10^5 \times 540)$

$= 985$ K

(ii) $\Delta U = Q + W$

Since Q =0, gas compress, W is positive. Thus ΔU must increases.

ΔU increases, kinetic energy of the atoms also increases.

KE of atoms is proportional to thermodynamic temperature, i.e KE =3/2kT

KE increase, thus temperature must increase.

6(c)

	U	*q*	*w*
the compression of an ideal gas at constant temperature	0	-	+
the heating of a solid with no expansion	+	+	0
The melting ice at 0 ◦C to give water at 0 ◦C. (Note: ice is less dense than water)	+	+	+

7(a)(i) When the wavelength of the wave is comparable to slit width, diffraction is most